CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
Seventh Semester of B. Tech. Examination (EC)
November 2013
EC 402RF & Microwave Engineering

Date: 14.11.2013, Thursday **Time: 10.00a.m. To 01:00 p.m.** **Maximum Marks: 70**

Instructions:
1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

Q-1 Choose appropriate option for question (a) to (g)

(a) Below the cut-off frequency (waveguide), there is ______________. 1
(a) Attenuation (b) No attenuation(c)Amplification(d)None of these.

(b) In the TEM mode, E_z ______ and H_z =_______ 1
(a)0, 0(b) 0, ∞(c) ∞, 0 (d)None of these

(b) Transmission lines facilitate _________propagation of energy. 1
(a)Guided(b) Unguided(c) None of these (d) Cannot say

(d) The reflection coefficient can have any value between 1
(a)0 and 1 (b)0 and infinity (c) -1 and +1(d) d)1 and infinity

(e) A transmission line is distortion less if 1
(a) RL=1/GC (b) RL=GC (c) LG=RC (d)RG=LC

(f) In the TM mode, E_z _____ and H_z = ________. 1
(a)0, 0(b)0, ∞(c) ∞ 0 (d) None of these

(g) Many circles are drown in a smith chart used for transmission line calculations. The circles are shown in Fig 1 presents 1
a) unit circle
b) constant reactance circle
c) constant resistance circle
d) constant reflection circle

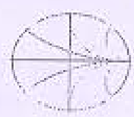

Figure 1

(h) Why TEM mode cannot propagate in Waveguide 2

(i) Explain standing wave in brief. 2

Q -2

(a) Derive the wave equation with complete solutions regarding wave propagation in rectangular waveguide for TM mode 10

(b) An air filled hollow rectangular waveguide has cross sections of $7X3.5cm^2$. Calculate cutoff frequency for dominant mode? 2

OR

Q-2

(a) Use Smith chart to find the following quantities for the transmission-linecircuit shown in the Fig 2. 10
(a) The SWR on the line.
(b) The reflection coefficient at the load.

(d) The input impedance of the line.
(e) The distance from the load to the first voltage minimum.
(f) The distance from the load to the first voltage maximum.

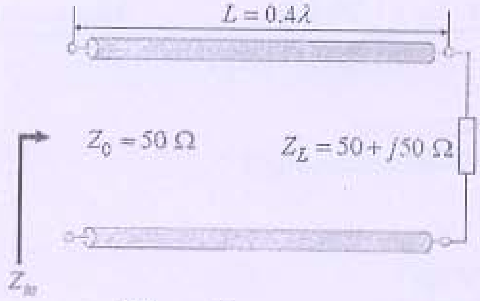

Figure 2

(b) Design a quarter wave transmission transformer to match a lone having impedance of 200 ohm ,feeding at 30MHz to a load of 800ohm — 2

Q-3
(a) Based on distribution theory of transmission line, draw equivalent circuit of transmission line and equation of voltage and current for lossless transmission line. Discuss standard assumptions. — 6
(b) A rectangular air filled waveguide with dimension 2.5X5 cm^2 cross is operated with a dominant mode with signal wavelength 4.5 cm. Find guided wavelength, phase velocity, phase constant — 6

OR

Q-3
(a) Explain faraday rotation principle and faraday rotation isolator in detail — 6
(b) Explain operation of magic tee along with S parameters — 6

Section –II

Q-4 Choose appropriate option for (a) to (f)
(a) Hollow metal conducting pipes designed to carry and constrain the electromagnetic waves of a microwave signal are — 1
(a)wave trap(v)waveguide(c)TWTA(d)microwave tube
(b) A microwave component which is used to interconnect two sections of waveguide is the — 1
(a)T section(b)bend(c)tapered section(d)choke point
(c) A waveguide like device that acts as a high-Q resonant circuit is a — 1
(a)microstrip (b)klystron(b)horn (d) cavity resonator
(d) A three-port microwave device used for coupling energy in only one direction around a closed loop is a — 1
(a)isolator(b)circulator(c)joint(d)terminator
(e) Which of the following diodes is not typically used in the microwave region? — 1
(a)point contact(b) standard PN(c)hot carrier (d)schottky
(f) Which of the following diodes does not oscillate due to negative-resistance characteristics? — 1
(a)tunnel(b)gun(c)SCR(d)IMPATT

(g)	Why strapping is required in magnetron?	1
(h)	What structure in the TWT delays the forward progress of the traveling wave?	1
(i)	Explain electron bunching process with help of diagram	3
Q-5		
(a)	A reflex klystron is designed for a frequency of 9GHz.The dc beam voltage is 600V for 1 ¾ mode and the distance from cavity to replier is 1 mm, dc beam current 10mA.Calculate repeller voltage efficiency and output power .Beam coupling coefficient is 1.	6
(b)	A microstripline has following parameters ε_r=5.23,h=0.1778mm,t=0.07112mm,w=0.254mm,calculate z_0	2
(c)	A transmission line has characteristic impotence of 50+j.01 ohm and is terminated in a load impedance of 73-j42.5 ohm. Calculate reflection coefficient and standing wave ration.	4

OR

Q- 5		
(a)	Explain Parametric up-converter and parametric down-converter with the help of equations.	6
(b)	What is tunneling? Explain energy band diagram and IV characteristics of tunnel diode.	6
Q-6		
(a)	Explain directional coupler in detail	6
(b)	Explain waveguide bend, twist and corner with the help of diagram	6

OR

Q- 6		
(a)	Draw appropriate diagram of TWTA and explain amplification process along with diagram.	6
(b)	Explain microstrip line and strapline in brief	6